Paths to Groups: Homotopy, Coverings, Fundamental Groups, and Presentations

N-20221223

§1.1 Path concatenation is reversible but not strictly associative

A path in a space X is a continuous map

$$\gamma : [0,1] \to X.$$

Its initial and terminal points are

$$s(\gamma) = \gamma(0), \qquad t(\gamma) = \gamma(1).$$

If $t(\gamma) = s(\eta)$, define the concatenation

$$(\gamma * \eta)(u) = \begin{cases} \gamma(2u), & 0 \le u \le \frac{1}{2}, \\ \eta(2u-1), & \frac{1}{2} \le u \le 1. \end{cases}$$

The reverse path is

$$\overline{\gamma}(u) = \gamma(1-u).$$

At the level of parametrized paths, concatenation is not strictly associative. The paths

$$(\alpha * \beta) * \gamma \qquad \text{and} \qquad \alpha * (\beta * \gamma)$$

traverse the same geometric pieces but allocate different subintervals of $[0,1]$ to them. Likewise, $\gamma * \overline{\gamma}$ is not literally the constant path. The desired algebra appears only after quotienting by a suitable deformation relation.

§1.2 Homotopy relative endpoints is a congruence for concatenation

Two paths $\gamma_0, \gamma_1 : [0,1] \to X$ with the same endpoints are homotopic relative endpoints if there is a continuous map

$$H : [0,1] \times [0,1] \to X$$

such that

$$H(u,0) = \gamma_0(u), \qquad H(u,1) = \gamma_1(u),$$

and

$$H(0,v) = s(\gamma_0), \qquad H(1,v) = t(\gamma_0)$$

for every v.

This is an equivalence relation. It is also compatible with concatenation: if

$$\gamma_0 \simeq \gamma_1 \text{ rel } \partial I, \qquad \eta_0 \simeq \eta_1 \text{ rel } \partial I,$$

then

$$\gamma_0 * \eta_0 \simeq \gamma_1 * \eta_1 \text{ rel } \partial I.$$

Indeed, concatenate the two homotopies at each time v. Thus homotopy relative endpoints is a congruence for path composition.

Reparametrization through an increasing map $r : [0,1] \to [0,1]$ with $r(0) = 0$, $r(1) = 1$ does not change the homotopy class. A linear interpolation between $r(u)$ and u gives the homotopy. This removes the associativity defect:

$$[\alpha] * ([\beta] * [\gamma]) = ([\alpha] * [\beta]) * [\gamma].$$

Similarly,

$$[\gamma] * [\overline{\gamma}] = [c_{s(\gamma)}], \qquad [\overline{\gamma}] * [\gamma] = [c_{t(\gamma)}].$$

§1.3 The fundamental groupoid remembers every basepoint at once

The fundamental groupoid $\Pi_1(X)$ has

- points of X as objects;
- homotopy classes relative endpoints of paths as morphisms;
- concatenation as composition.

Every morphism is invertible, with inverse represented by the reversed path. This is why a groupoid, rather than a monoid, is the natural first algebraic object attached to a space.

Fixing a basepoint x_0 and taking endomorphisms gives the fundamental group

$$\pi_1(X, x_0) = \mathrm{Aut}_{\Pi_1(X)}(x_0).$$

Its elements are based loops modulo based homotopy.

If ρ is a path from x_0 to x_1, basepoint change is

$$\rho_\# : \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad [\gamma] \longmapsto [\overline{\rho} * \gamma * \rho].$$

A different choice of ρ changes this isomorphism by an inner automorphism. The groupoid keeps this dependence visible instead of hiding it behind a noncanonical identification.

§1.4 The exponential covering turns loops into integer endpoints

Consider

$$p : \mathbb{R} \to S^1, \qquad p(t) = e^{2\pi i t}.$$

For each $t_0 \in \mathbb{R}$, restriction to

$$\left(t_0 - \frac{1}{2}, t_0 + \frac{1}{2}\right)$$

is a homeomorphism onto $S^1 \setminus \{-p(t_0)\}$. Thus p is a covering map.

Let $\gamma : [0,1] \to S^1$ and choose $a \in \mathbb{R}$ with $p(a) = \gamma(0)$. Subdivide $[0,1]$ so that the image of each subinterval lies in one evenly covered arc. On the first subinterval use the inverse branch whose value at 0 is a; on each later subinterval choose the unique branch agreeing at the common endpoint. This constructs a unique lift

$$\widetilde{\gamma} : [0,1] \to \mathbb{R}, \qquad p \circ \widetilde{\gamma} = \gamma, \qquad \widetilde{\gamma}(0) = a.$$

If γ is a loop based at 1 and $\widetilde{\gamma}(0) = 0$, then

$$e^{2\pi i \widetilde{\gamma}(1)} = 1,$$

so

$$\widetilde{\gamma}(1) \in \mathbb{Z}.$$

Define the winding number by

$$\mathrm{wind}(\gamma) = \widetilde{\gamma}(1).$$

Concatenation adds endpoints of lifts:

$$\mathrm{wind}(\gamma * \eta) = \mathrm{wind}(\gamma) + \mathrm{wind}(\eta).$$

A based homotopy of loops lifts after fixing the initial lift. The terminal lift is a continuous integer-valued function of the homotopy parameter, hence constant. Therefore winding number depends only on the based homotopy class.

§1.5 The covering computation gives $\pi_1(S^1) \cong \mathbb{Z}$

The map

$$W : \pi_1(S^1, 1) \to \mathbb{Z}, \qquad W([\gamma]) = \mathrm{wind}(\gamma)$$

is a homomorphism.

It is surjective because

$$\gamma_n(u) = e^{2\pi i n u}$$

has lift $\widetilde{\gamma}_n(u) = nu$ and winding number n.

It is injective. Suppose $W([\gamma]) = 0$. The lift $\widetilde{\gamma}$ is then a loop in $\mathbb{R}$ based at 0. The linear homotopy

$$\widetilde{H}(u,v) = (1-v)\widetilde{\gamma}(u)$$

contracts it to the constant loop. Projecting by p gives a based null-homotopy of γ. Hence

$$\ker W = 0.$$

Therefore

$$\boxed{\pi_1(S^1, 1) \cong \mathbb{Z}.}$$

The group operation is addition of winding numbers, and reversal changes the sign.

§1.6 The punctured plane retracts onto the circle

Let

$$\mathbb{C}^* = \mathbb{C} \setminus \{0\}.$$

The map

$$r : \mathbb{C}^* \to S^1, \qquad r(z) = \frac{z}{|z|}$$

is a deformation retraction. Define

$$H(z,v)=\left((1-v)+\frac{v}{|z|}\right)z.$$

The scalar coefficient is positive, so $H(z,v)\neq 0$. Moreover,

$$H(z,0)=z,\qquad H(z,1)=\frac{z}{|z|},$$

and if $|z|=1$, then $H(z,v)=z$ for all v. Thus the inclusion $S^1\hookrightarrow\mathbb{C}^*$ and r are homotopy inverses.

Consequently,

$$\boxed{\pi_1(\mathbb{C}^*,1)\cong\mathbb{Z}.}$$

The missing point is measured exactly by winding number.

§1.7 Continuous maps induce homomorphisms

A based continuous map

$$f:(X,x_0)\to(Y,y_0)$$

sends a loop γ to $f\circ\gamma$. It preserves homotopies and concatenation, hence induces

$$f_*:\pi_1(X,x_0)\to\pi_1(Y,y_0),\qquad f_*([\gamma])=[f\circ\gamma].$$

Functoriality is immediate:

$$(g\circ f)_*=g_*\circ f_*,\qquad (\mathrm{id}_X)_*=\mathrm{id}_{\pi_1(X)}\,.$$

For

$$f_m:S^1\to S^1,\qquad f_m(z)=z^m,$$

one has

$$f_m\circ\gamma_n(u)=e^{2\pi imnu}.$$

Under $\pi_1(S^1)\cong\mathbb{Z}$, the induced homomorphism is therefore

$$(f_m)_*:\mathbb{Z}\to\mathbb{Z},\qquad n\longmapsto mn.$$

This computation distinguishes the maps $z\mapsto z^m$ up to based homotopy.

§1.8 Van Kampen assembles groups from overlapping pieces

Let $X=U\cup V$, where $U,V,U\cap V$ are path-connected open sets containing a basepoint x_0. The Seifert–van Kampen theorem states that the diagram

$$\begin{tikzcd}
\pi_1(U\cap V,x_0) \arrow[r] \arrow[d] & \pi_1(U,x_0) \arrow[d] \\
\pi_1(V,x_0) \arrow[r] & \pi_1(X,x_0)
\end{tikzcd}$$

is a pushout in the category of groups. Concretely,

$$\pi_1(X,x_0)\cong\frac{\pi_1(U,x_0)*\pi_1(V,x_0)}{\langle\!\langle i_{U*}(\omega)i_{V*}(\omega)^{-1}:\omega\in\pi_1(U\cap V,x_0)\rangle\!\rangle}.$$

The free product records loops from the two pieces; the normal closure imposes the identifications already visible in the overlap.

For the figure-eight

$$X = S^1 \vee S^1,$$

choose neighborhoods retracting onto the two circles whose intersection is contractible. The overlap contributes no relation, so

$$\boxed{\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} = F(a,b).}$$

The words

$$ab, \quad ba, \quad aba^{-1}b^{-1}$$

represent genuinely different loop classes; no commutativity relation is present.

§1.9 Attaching a two-cell adds one normal relation

Let X be path-connected and attach a disk along a loop

$$\varphi : S^1 \to X.$$

The resulting space

$$X \cup_\varphi D^2$$

contains a new disk across which the attaching loop can contract. Van Kampen gives

$$\pi_1(X \cup_\varphi D^2) \cong \pi_1(X)/\langle\langle[\varphi]\rangle\rangle.$$

Thus cell attachment translates directly into a group presentation.

A torus has a CW structure with one vertex, two one-cells a, b, and one two-cell attached along

$$aba^{-1}b^{-1}.$$

Starting from the figure-eight gives

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} = 1\rangle \cong \mathbb{Z}^2.$$

The two basic loops commute because the square filling of the commutator becomes the two-cell of the torus.

§1.10 Connected coverings of the circle are indexed by subgroups of $\mathbb{Z}$

For each $n \geq 1$, the map

$$p_n : S^1 \to S^1, \qquad p_n(z) = z^n$$

is an n-sheeted connected covering. Its induced subgroup is

$$(p_n)_*\pi_1(S^1) = n\mathbb{Z} \subseteq \mathbb{Z}.$$

Conversely, every subgroup of $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n \geq 0$. The general classification theorem for sufficiently well-behaved spaces says that connected based coverings correspond to subgroups of the fundamental group. For the circle, the entire classification is already visible in the elementary maps $z \mapsto z^n$.

The progression is now concrete:

$$\text{paths} \longrightarrow \text{homotopy classes} \longrightarrow \text{groupoid and groups} \longrightarrow \text{coverings and presentations.}$$

Algebra enters because deformation classes of reversible paths carry composition, and topology becomes computable because coverings and gluings turn that composition into integers, free words, and relations.